INCH-POUND

MIL-PRF-26542/12A
30 May 1997
SUPERSEDING
MIL-M-26542/12
29 March 1985

PERFORMANCE SPECIFICATION SHEET

MICROPHONE ASSEMBLY, M-169A/AIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-26542.

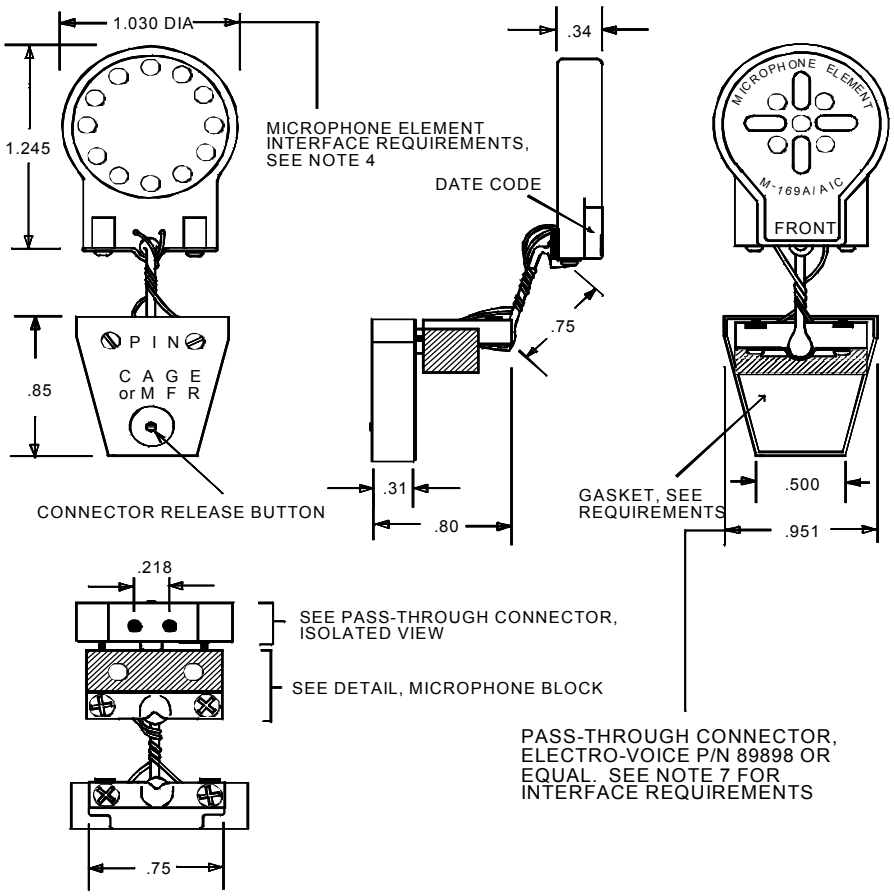
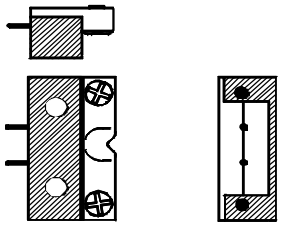


FIGURE 1. Microphone assembly M-169A/AIC

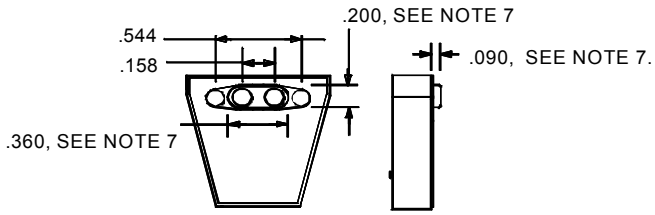
Ⓐ denotes changes.

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Ⓐ DETAIL, MICROPHONE BLOCK, RECOMMENDED DESIGN, SEE REQUIREMENTS (BLOCK)



Ⓐ PASS-THROUGH CONNECTOR, ISOLATED VIEW

NOTES:

1. Dimensions are in inches. Tolerance is ±.015 inches, unless otherwise specified.
2. Quantity and configuration of sound ports is optional.
3. Location of marking on sides shown is optional.
4. Dimensions of microphone element shall provide interface to the dimensions of the interior space in the mask facepiece, allowing the element to position closely to the user's mouth without becoming obstructed by other mask components, such as valve terminals.
5. Dimensions of the microphone block shall provide interface to the pass-through holes of the mask facepiece, as well as to the exterior bracket, and shall ensure a leak-proof seal.
6. All non-dimensioned contours and illustration components shall be designed to meet the performance requirements of this specification and MIL-PRF-26542.
7. Dimensions of the pass-through connector shall provide interface to electrical connectors, shall provide a leak-proof seal at the microphone pin openings, and shall allow a firm seating of the microphone block against the interior of the mask facepiece when tightly installed. The pass-through connector dimension 0.090 is intended for interface to the corresponding reduced thickness region of the MBU-20/P (rubber) facepiece, as well as to the depth of the MBU-12 hardshell opening, to provide a firm seal and secure electrical connection.
8. When approved by the qualifying activity, the microphone block may be supplied with additional removable material for preventing damage to the facepiece, such as a gasket. Any gasket must be thin enough to allow a secure electrical connection of the block to the pass-through connector. In addition, it must be constructed of a durable material capable of withstanding use for at least 30 days (1 cleaning cycle), and meet the Unsafe Materials requirement (see below).

FIGURE 1. Microphone assembly M-169A/AIC - Continued.

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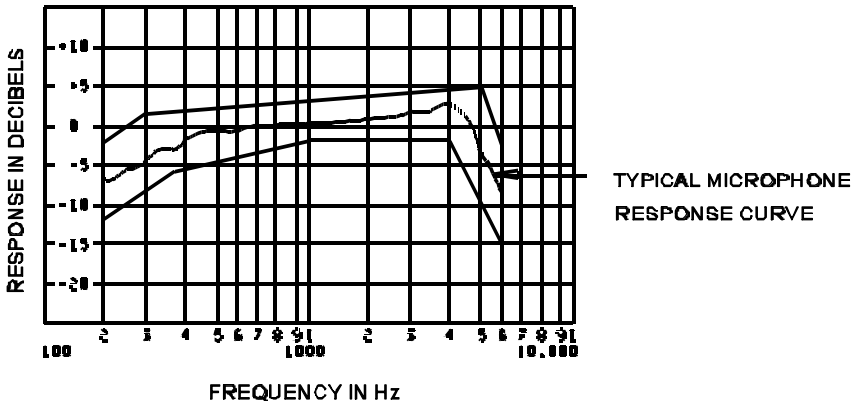


FIGURE 2. Frequency response at sea level

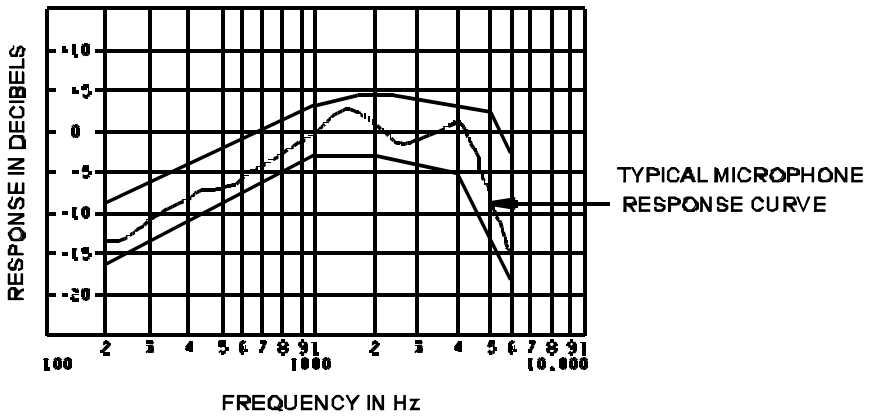


FIGURE 3. Frequency response at 25,000 feet

REQUIREMENTS:

Component parts: Shall be in accordance with figure 1.

Release button: Shall be recessed to prevent obstruction and inadvertent release of connector.

Bracket rod: Shall be bent to 90±2 deg at the microphone-element end, as shown in figure 1, to provide a complete range of adjustability of the element when installed in the oxygen mask. The rod shall be built from a high-strength, corrosion-resistant material meeting the requirements of MIL-PRF-26542, such as stainless-steel.

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Microphone block:

Screw-receiving sections: The material which receives the mask-installation screws (see figure 1, “detail, microphone bracket”, shaded portions) shall prevent breakage and screw slippage, having a single metal section enclosing and connecting the two screw-receiving sections. The metal shall be either nickel-plated brass, or another material with equivalent or superior strength, corrosion-resistance, and adherence to the requirements of MIL-PRF-26542. The design shown in figure 1 (detail, microphone block) shall be used, or a design which meets or exceeds its performance without introducing mask damage or other hazards to the user, upon approval from the qualifying activity.

Interface to oxygen-mask material: The bracket shall be designed to prevent damage to the oxygen mask material to which it secures in-use. This shall include damage resulting both from: (a) torque stresses on the element resulting from pilot-imposed forces such as adjustments to the microphone element orientation, or pull-forces from associated cable and hose assemblies during normal head motion, and (b) possible compression of the mask face-piece due either to altitude pressure changes, or intentional squeezing of the nose-section as a result of the periodic need to clear sinus cavities during use.

Pass-through connector: The connector shall include a gasket material, covering the mask-interface surface, which provides an air-tight, slip-resistant seal, such as silicon rubber, meeting the performance requirements of MIL-PRF-26542.

Unsafe materials: Parts and adhesives which install inside the mask, and which may become detached during normal use and thus create an ingestion hazard, or which otherwise present a chemically toxic hazard to the user, shall not be used.

Hardware: The product shall be supplied with 2 stainless steel screws for securing the pass-through connector to the microphone block. The screws shall be type 2-56 X ½ fillister head, slotted, for interchangeabilityof spares.

Color: In accordance with MIL-PRF-26542, with the exception that metal components shall be not have surface coloring, but shall retain their natural appearance.

Weight: 22 grams, maximum.

Performance:

Sensitivity at ground level: 34.96 dB - 40.98 dB (re 1 µV) or 55.97 µV - 111.94 µV with an SPL input of 28 dynes/cm^2 at 1 kHz.

Sensitivity at altitude: At a simulated altitude of 25,000 feet, the sensitivity shall be equal or not more than 8 dB of the ground level sensitivity.

Frequency response: The frequency response at ground level shall be within the limits shown (figure 2), and the frequency response at 25,000 ft shall be within the limits shown (figure 3). The frequency response range shall be 200-6000 Hz.

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Impedance: 4.0 ohms to 6.0 ohms. The impedance at any frequency over the range of 200 Hz to 6,000 Hz shall not deviate from the 1,000 Hz impedance by more than 20 percent.

Resistive load: 5.0 ohms.

Dielectric withstanding voltage: This test may be performed on the bracket in isolation from the microphone element, at the option of the manufacturer.

Microphone leakage: There shall be no more than 100 ml/min of leakage, and the test shall only be performed at 60,000 ft simulated altitude. This test shall be performed at a laboratory acceptable to the Qualifying Activity.

Intended use. M-169A/AIC is a dynamic, noise-canceling moving coil microphone designed for use in an oxygen mask or pressure-type oxygen helmet, at altitudes where the use of an oxygen helmet is required. The microphone element is intended for use with intercommunication sets AN/AIC-10( ) and AN/AIC-18. The microphone is used with both the MBU-12 and MBU-20/P (Combat Edge) oxygen masks.

The microphone assembly shall be tested in accordance with the tests listed in table II.

TABLE I. Part Number Designations

Part or Identifying Number (PIN)	Characteristics
M-169A/AIC	Microphone assembly as shown above.
M26542/12-01	Pass-through connector, alone.

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Ⓐ TABLE II. Parameter applicability

Inspection	Requirement paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1, 3.4, 3.5	4.5.1	X	X		
Sensitivity at ground level	3.5.1	4.5.2.2	X	X		
Sensitivity at altitude	3.5.2	4.5.2.3	X			
Frequency response at ground leve	3.5.3	4.5.2.4	X	X		
Frequency response at altitude	3.5.4	4.5.2.5	X			
Impedance	3.5.5	4.5.3	X	X		
Noise cancellation characteristics	3.5.7.1	4.5.5.1	X			
Effect of external magnetic field	3.5.8	4.5.6	X			
Stray magnetic field	3.5.9	4.5.7	X			
Linearity	3.5.10	4.5.8	X			
Talk-out	3.5.11	4.5.9	X	X		
Dielectric withstanding voltage	3.5.12	4.5.10	X			
Signal-to-noise	3.5.7	4.5.5	X		X	
Distortion	3.5.6	4.5.4	X		X	
Interchangeability	3.5.13	4.5.11	X		X	
Group II						
Thermal shock	3.5.14	4.5.12	X			X
Humidity	3.5.15	4.5.13	X			X
Drop	3.5.16	4.5.14	X			X
Pressure equalization	3.5.17	4.5.15	X			X
Explosive decompression	3.5.18	4.5.16	X			X
Salt fog	3.5.19	4.5.17	X			X
Group III						
Vibration	3.5.20	4.5.18	X			X
Bounce	3.5.21	4.5.19	X			X
Altitude	3.5.22	4.5.20	X			X
Moisture barrier seal	3.5.23	4.5.21	N/A			N/A
Immersion	3.5.24	4.5.22	N/A			N/A
Group IV						
Fungus	3.5.25	4.5.23	X			
Group V						
Gun blast	3.5.26	4.5.24	N/A			
Microphone leakage	3.5.27	4.5.25	X			

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME
Navy - AS, OS
Air Force - 31, 99
DLA - CC

Preparing activity:
Air Force - 85

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